Table 4. Symmetry classification of the 2D and 3D isotropic harmonic oscillators for the corresponding vibrational angular momentum index L_2 and the multiplicity index M_i ($i = 3, 4$, $0 \leq M_i \leq L_i$), respectively.

L_2	Symmetry 2D	M_i	Symmetry 3D
0	A_1	0	A_1
$3n$	A_1, A_2	$12n + 0$	A_1
$3n \pm 1$	E	$12n + 6$	A_2
		$6n \pm 2$	E
		$4n + 3$	F_1
		$4n + 1$	F_2

n is an arbitrary integer, $n = 1, 2, 3, \ldots$.

$\ldots, 0(1)$. $M_i \leq L_i$ is a multiplicity index used to count states within a given n_i, L_i set, see Boudon, Rey & Loëte (2006). The advantage of this scheme is that the angular momentum indices L_i (for the E-type ν_2 mode) and M_i (for the F-type ν_3 and ν_4 modes) provide a one-to-one correspondence to the symmetry of the corresponding component as listed in Table 4. In fact, such a correspondence is common for the 2D E-type symmetry isotropic harmonic oscillators, see e.g. Bunker & Jensen (1998) and Yurchenko et al. (2005), while the 3D isotropic harmonic oscillator quantum numbers (Hougen 2001) are less common.

The normal mode assignment is in fact reconstructed using the local mode quantum numbers. The easiest way to correlate them is at step 2 of the TROVE calculations, when the three reduced eigenvalue problems in equations (21)–(23) are solved. Indeed, the local mode eigensolutions of the 4D Hamiltonian $\hat{H}^{(1)}$ in equation (21) need to match the combinations of stretching 1D and 3D harmonic oscillators only; the eigenvalues of the 2D bending $\hat{H}^{(2)}$ are easily matched to energies of a 2D isotropic harmonic oscillator; and the eigenvalues of the 3D bending reduced problem for $\hat{H}^{(3)}$ are then correlated solutions of a 3D harmonic oscillator. Once the correlation is established, it is propagated through the multistep assignment procedure along with the local mode quantum numbers. Moreover, the unique vibrational state index $i_{\rm vib}$ can also be used to efficiently re-assign the TROVE local mode quantum numbers to any other quantum numbers scheme, which is possible even in the final data set.

The CH_4 experimental quantum number set usually consists of J, $\Gamma_{\rm tot}$, polyad number P in equation (19) and a counting number n, counting different states within the same J, $\Gamma_{\rm tot}$, and P. It is an extremely simple, and therefore powerful, set for correlating methane spectroscopic data from different sources. It has been adopted by the community as the methane spectroscopic standard and it was used to classify the experimental data in the recent MARVEL work on CH_4 (Kefala et al. 2024). The disadvantage of the experimental quantum number scheme is that it loses any connection with the physics, such as, e.g. vibrational state, which can be required, e.g. when it is important to know if a given state has a bending or stretching nature, whether it belongs to a forbidden or combination band, and so on.

An important aspect of the ExoMol States file is the state degeneracies $g_i = g_{\rm ns}(2J + 1)$. For $^{12}CH_4$, considering the H nuclear spin, the nuclear statistical weight, $g_{\rm ns}$, takes the symmetry-dependent values 5, 5, 2, 3, and 3 for A_1, A_2, E, F_1, and F_2, respectively.

The electric dipole transitions obey the following symmetry-determined selection rules:

$$A_1 \leftrightarrow A_2 \ , E \leftrightarrow E \ , F_1 \leftrightarrow F_2 \tag{28}$$

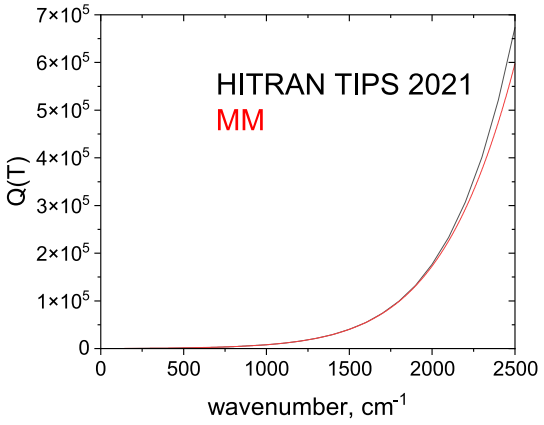


Figure 4. Partition functions of CH_4: HITRAN TIPS 2021 (Gamache et al. 2021) and MM.

with the standard rotational angular momentum, **J**, selection rules:

$$J \leftrightarrow J \pm 1 \ , J' + J'' \neq 0. \tag{29}$$

5.2 Partition function

A partition function of CH_4 computed using the MM line list is provided as part of the line list package, covering the temperature range from 0 to 2500 K on a grid of 1 K. In Fig. 4, the MM partition function is compared to the HITRAN partition function generated using the TIPS 2021 package (Gamache et al. 2021). The MM partition function closely agrees with TIPS until about $T = 2000$ K, after which it slowly deviates, underestimating the more complete TIPS partition function. Hence, $T = 2000$ K is our estimated upper temperature limit for completeness of the MM line list.

5.3 Opacities

Temperature- and pressure-dependent opacities of CH_4 have been generated using the ExoMolOP procedure (Chubb et al. 2021) for the MM line list. These are provided for four exoplanet atmospheric retrieval codes ARCiS (Min et al. 2020), TauREx (Al-Refaie et al. 2021), NEMESIS (Irwin et al. 2008), and petitRADTRANS (Molliére et al. 2019). For the line broadening, we assumed an 85 per cent H_2 and 15 per cent He atmosphere and Voigt line profile with the following parameters: $\gamma_{\rm H_2} = 0.12$ cm^{-1}, $n_{\rm H_2} = 0.5$, $\gamma_{\rm He} = 0.05$ cm^{-1}, and $n_{\rm He} = 0.5$.

6 SPECTROSCOPIC ILLUSTRATIONS

We use the MM line list to simulate spectra of CH_4 to demonstrate the accuracy of the line positions, the quality of the line intensities, as well as its completeness especially at high temperatures. To this end, we compare simulated spectra either with experimental or theoretical spectra from the literature or other data bases.

We start with a comparison against the HITRAN spectrum of $^{12}CH_4$ at $T = 296$ K. Figs 5 and 6 provides detailed comparisons of MM and HITRAN 'stick' spectra at $T = 296$ K, where absorption coefficients (cm per molecule) are plotted as sticks as functions of the wavenumber covering the region between 2700 and 12000 cm^{-1}. There is very good agreement between MM and HITRAN below 6200 cm^{-1}, where all the lines shown are MARVELized, also nicely illustrating the completeness of the CH_4 MARVEL energies in this

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